

A Method for Computing Semantics from Physical Objects and their Governing Laws (*Theoretical Paper*)

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Abstract

This work shows how physical scene descriptions may correspond with semantic vectors and categories. Scenes and objects have fundamental field descriptions, distinguished by volume. Strong assumptions around object interactions and observer-strain deformations allow the construction of semantic vector spaces and discrete classes. Under these assumptions, geometry and object-interactions alone can yield tractable information-theoretic representations of reality that one might consider semantic.

1 Introduction

1.1 Abridged Setup

Consider the physical and informational universes (P, I) discursively and analytically distinct, in progenitor-agnostic parallel.

Definition 1 (Physical and Informational Universes). Let P and I denote the collection of physically instantiated scenes and informational representations, respectively. The distinction between P and I is analytic, not ontologically directional: make no assumption that either collection generates the other.

Denote a correspondence between a physical scene description and equivalent informational description $h \in H : I \rhd P \leftrightarrow I$. This study seeks one such correspondence between field-theoretic scenes and their semantic vector spaces and categorisations.

Assumption 1.1 (Physical Closure of the Observer). Every observer, probe, initial state, and interaction process that the construction uses exists within the complete physical scene $\Sigma \in P$. h receives no external semantic labels, target partitions, or non-physical observer variables.

Assumption 1.2 (Fixed Descriptive Specification). Fix a descriptive specification

$$\Sigma_{\text{desc}} := (\varepsilon, \mathcal{N}, [\underline{\phi}, \overline{\phi}], \mathfrak{A}), \quad (1)$$

where $\varepsilon > 0$ is the descriptive resolution, $\mathcal{N}$ the field-normalisation convention, $[\underline{\phi}, \overline{\phi}]$ the retained field range, and $\mathfrak{A}$ the coordinate convention used to represent the fields. All comparisons within a construction use the same Σ_{desc} . The coordinates in $\mathfrak{A}$ are a representational choice, not privileged physical coordinates. The author addresses coordinate covariance separately.

Conjecture 1.3 (Physical Sufficiency for Observer-Relative Meaning). *For every physically closed scene Σ there exists an assignment*

$$\check{h}(\Sigma) = (V_\Sigma, g_\Sigma, \mathcal{C}_\Sigma), \quad (2)$$

where V_Σ is a smooth observer-relative representation space, g_Σ is V_Σ 's observer-relative geometry, and

$$\mathcal{C}_\Sigma \in \text{Part}(V_\Sigma) \quad (3)$$

V_Σ 's observer-relative categorisation.¹

1.2 Main Results

Theorem 1.4 (Local Physical Recovery of Observer-Relative Meaning). *Let Σ be a physically closed scene described under a fixed Σ_{desc} . Suppose Σ 's: (i) observation windows belong to the regular window domain; (ii) quotient fibres form the regular s -catalogue; (iii) observer-response map is smooth and regular; (iv) physical response-equivalence relation is well-defined and stable.*

Then, locally around every regular observed scene, Σ determines: (i) a smooth vector-space representation; (ii) an observer-relative pullback geometry; and (iii) a well-defined unlabelled partition into observer-relative categories.

*These objects are unique up to smooth coordinate change and permutation of the category cells.*²

Theorem 1.5 (No Assumption-Free Recovery). *Let X be any geometric information supplied to a discretisation method, and let Λ_X be the collection of admissible assumption objects compatible with X . For each $\Lambda \in \Lambda_X$, let*

$$\mathcal{C}_\Lambda^* \in \text{Part}(V) \quad (4)$$

denote the corresponding target partition.

If there exist $\Lambda_1, \Lambda_2 \in \Lambda_X$ such that

$$\mathcal{C}_{\Lambda_1}^* \neq \mathcal{C}_{\Lambda_2}^*, \quad (5)$$

then no discretisation method

$$\mathbf{D} : X \longrightarrow \text{Part}(V) \quad (6)$$

depending only on X can guarantee recovery of $\mathcal{C}_\Lambda^$ for every $\Lambda \in \Lambda_X$.*

Proof. Because $\mathbf{D}$ receives the same input X under both Λ_1 and Λ_2 , it must return the same partition in both cases. That partition cannot equal two distinct target partitions. Hence $\mathbf{D}$ fails for at least one admissible assumption object. $\square$

¹The assignment $\check{h}(\Sigma)$ depends only on the physical scene Σ and the fixed descriptive specification. $\check{h}(\Sigma)$ is invariant under the physical-scene isomorphisms defined in Appendix ?? Physically isomorphic scenes must therefore produce smoothly isomorphic representation spaces and equivalent partitions, up to the relabelling of their cells.

²Section ?? assembles Theorem 1.4's proof.

2 Current Literature

The literature documents a long history of works deriving informational assignments from descriptions of physical objects. Atiyah and Singer (1968) proved continuous differential operators relate to discrete integer invariants [3]. In Topological and Geometric Deep Learning (TDL/GDL), researchers have mapped object-data to identifiers like Betti numbers [4], and physical scenes to classifications via general geometry [5].

Earlier works typically captures a local projection of the problem. Computer vision and open-vocabulary 3D mapping connect sensed scenes or reconstructed fields to semantic features [6], [7], [8], [9], [10]. TDL/GDL formalise learning over topological domains [4] and symmetry-aware representation [5]. Shape analysis formalises deformation and identity [11]; density geometry formalises physical configuration spaces through diffeomorphism groups and spaces of densities [12]; shape calculus studies smoothly moving observation domains [13]; and bisimulation, quotient learning, and quotient generative modeling formalise when detailed states may collapse without losing relevant structure [14], [15], [16], [17]. Symbol-grounding theory motivates the world-to-meaning bridge [18]. What remains for the current purpose is to organise these fragments into one physical-informational pipeline.

The contribution of this work therefore lies not in the novelty of its parts, but in their synthesis. This work seeks to show that a physically constrained field theoretic scene description can yield observer-relative, response-induced semantic geometry and categories in a way that is principled and computable throughout the physical domain. Since many of these technical ingredients already exist in neighbouring literatures³, the work’s main responsibility is aggregating these methods within the over-arching pipeline. Physical mass configurations, distinguishable object-hood, observer-relative meaning, and computable representations are unifiable. The current framing allows that unification and the comparison of these disparate components.

3 H Domain: Physical Objects

3.1 Physical Space Filtering via Field Descriptions

Definition 2 (Physical Domain). A physical domain is a pair

$$P := (M, g), \quad (7)$$

where M is a smooth, Hausdorff, second-countable, d -dimensional manifold and g is a smooth Riemannian metric on M . When required, M is assumed oriented, with induced volume measure dV_g .

An alternative is the point-configuration space where out-of-bounds points are quotiented over.

Definition 3 (State-Field Bundle). Let

$$\pi_S : S \longrightarrow M \quad (8)$$

be a smooth real vector bundle admitting a fixed field decomposition

$$S = \bigoplus_{j \in \mathcal{J}} S_j, \quad (9)$$

³(e.g., quotient spaces, shape spaces, density geometry, geometric deep learning, semantic embeddings, and symbol grounding)

where each $S_j \rightarrow M$ represents one retained physical field component at the chosen descriptive scale.

Definition 4 (Ambient Configuration Space). The ambient configuration space is the space of smooth global S-sections:

$$\Gamma(P, S) := \Gamma^\infty(M, S) = \{s : M \rightarrow S : \pi_S \circ s = \text{id}_M, s \text{ is smooth}\}, \quad (10)$$

equipped with standard C^∞ compact-open Fréchet topology.

Definition 5 (Physically Admissible Configurations). Let

$$\Gamma_{\text{phys}}(P, S) \subseteq \Gamma(P, S) \quad (11)$$

denote the configurations satisfying whatever physical laws, conservation conditions, or boundary conditions are imposed by a particular physical theory.

The present construction is defined on the ambient space $\Gamma(P, S)$ and does not require membership in $\Gamma_{\text{phys}}(P, S)$. Any later physical restriction may be implemented by restricting the construction to this subset.

Definition 6 (Finite Field Filtration). For ordered indices $\phi \leq \bar{\phi}$, define the filtered state-field bundle

$$S_{\phi; \bar{\phi}} := \bigoplus_{j=\phi}^{\bar{\phi}} S_j. \quad (12)$$

Let

$$\pi_{\phi; \bar{\phi}} : S \longrightarrow S_{\phi; \bar{\phi}} \quad (13)$$

be the corresponding fibrewise linear projection. It induces the configuration-space filtration

$$\Pi_{\phi; \bar{\phi}} : \Gamma(P, S) \longrightarrow \Gamma(P, S_{\phi; \bar{\phi}}), \quad (14)$$

$$\Pi_{\phi; \bar{\phi}}(s) := \pi_{\phi; \bar{\phi}} \circ s. \quad (15)$$

Assumption 3.1 (Configuration-Space Regularity). All section spaces are treated as Hausdorff Fréchet spaces with their standard convenient smooth structures. Restriction maps, fibrewise field projections, and pullbacks by the smoothly varying bundle diffeomorphisms used below are assumed continuous linear in the section variable and smooth in their finite-dimensional parameters.

Assumption 3.2 (Fixed Resolution and Field Content). Within a single configuration space, the resolution ε , normalisation convention $\mathcal{N}$, and retained field range $[\phi, \bar{\phi}]$ fixed in Assumption 1.2 remain unchanged. Changes of resolution or field content therefore relate different configuration spaces rather than points within the same space.

Lemma 3.3 (Smoothness of Field Filtration). *The map*

$$\Pi_{\phi; \bar{\phi}} : \Gamma(P, S) \longrightarrow \Gamma(P, S_{\phi; \bar{\phi}}) \quad (16)$$

is smooth.

Proof. The bundle projection $\pi_{\phi; \bar{\phi}}$ is smooth and fibrewise linear. Its induced map on section spaces is continuous linear under Assumption 3.1. Every continuous linear map between the chosen locally convex spaces is smooth. $\square$

Lemma 3.4 (Compatibility of Successive Field Projections). *Let*

$$[\phi_2, \bar{\phi}_2] \subseteq [\phi_1, \bar{\phi}_1]. \quad (17)$$

Then

$$\Pi_{[\phi_1, \bar{\phi}_1] \rightarrow [\phi_2, \bar{\phi}_2]} \circ \Pi_{\phi_1; \bar{\phi}_1} = \Pi_{\phi_2; \bar{\phi}_2}. \quad (18)$$

Proof. Both sides retain exactly the field components indexed by $[\phi_2, \bar{\phi}_2]$ and discard every other component. The equality therefore holds fibrewise and hence on the induced section spaces. $\square$

3.2 Volumetric Distinguishability

Definition 7 (Parameterised Observation Windows). Let Θ be a finite-dimensional smooth parameter manifold and let

$$\alpha : \Theta \times M \longrightarrow \mathbb{R} \quad (19)$$

be smooth. For

$$a = (\theta, c) \in A := \Theta \times \mathbb{R}, \quad (20)$$

define

$$B_a = B_{\theta, c} := \{x \in M : \alpha_\theta(x) \geq c\}. \quad (21)$$

Definition 8 (Regular Parameter Domain). The regular parameter domain $A_{\text{reg}} \subseteq A$ is a chosen open set such that, for every $a = (\theta, c) \in A_{\text{reg}}$:

1. $B_{\theta, c}$ is non-empty; and
2. c is a regular value of α_θ , equivalently,

$$d_x \alpha_\theta \neq 0 \quad \text{for every } x \in \alpha_\theta^{-1}(c). \quad (22)$$

Definition 9 (Windowed Physical Object). For $s \in \Gamma(P, S)$ and $a \in A_{\text{reg}}$, the corresponding windowed physical object is

$$o_a(s) := (B_a, s|_{B_a}). \quad (23)$$

At this stage, objecthood means only a field configuration viewed within a specified observation window; no conventional semantic category is yet assigned.

Assumption 3.5 (Compactness and Properness). There exists a fixed compact domain $D \subseteq M$ such that

$$B_a \subseteq \text{int}(D) \quad \text{for every } a \in A_{\text{reg}}. \quad (24)$$

Define the total regular observation-window space

$$\mathcal{W}_{\text{reg}} := \{(a, x) \in A_{\text{reg}} \times D : x \in B_a\} \quad (25)$$

and its projection

$$\pi_{\mathcal{W}} : \mathcal{W}_{\text{reg}} \longrightarrow A_{\text{reg}}, \quad \pi_{\mathcal{W}}(a, x) = a. \quad (26)$$

The projection is assumed surjective. The compact-containment condition ensures that $\pi_{\mathcal{W}}$ is proper.

Lemma 3.6 (Borel Measurability of Regular Windows). *For every $a = (\theta, c) \in A_{\text{reg}}$,*

$$B_{\theta,c} \in \mathcal{B}(M), \quad (27)$$

where $\mathcal{B}(M)$ is the Borel σ -algebra of M .

Proof. The map $\alpha_\theta : M \rightarrow \mathbb{R}$ is continuous and $[c, \infty)$ is closed. Therefore

$$B_{\theta,c} = \alpha_\theta^{-1}([c, \infty)) \quad (28)$$

is closed in M , and hence Borel measurable. $\square$

Lemma 3.7 (Smoothness of the Total Observation-Window Space). *The space $\mathcal{W}_{\text{reg}}$ is a smooth manifold with boundary*

$$\partial\mathcal{W}_{\text{reg}} = \{(\theta, c, x) : \alpha_\theta(x) = c\}. \quad (29)$$

Proof. Define

$$F : A_{\text{reg}} \times D \longrightarrow \mathbb{R}, \quad F(\theta, c, x) := \alpha_\theta(x) - c. \quad (30)$$

Then

$$\mathcal{W}_{\text{reg}} = F^{-1}([0, \infty)), \quad \partial\mathcal{W}_{\text{reg}} = F^{-1}(0). \quad (31)$$

For every point of $F^{-1}(0)$, Definition 8 gives $d_x \alpha_\theta \neq 0$. Hence $dF \neq 0$ on $F^{-1}(0)$, so 0 is a regular value of F . The regular-level-set theorem therefore makes $F^{-1}([0, \infty))$ a smooth manifold with boundary $F^{-1}(0)$. $\square$

Lemma 3.8 (Boundary-Compatible Submersion). *Both*

$$\pi_{\mathcal{W}} : \mathcal{W}_{\text{reg}} \longrightarrow A_{\text{reg}} \quad (32)$$

and its boundary restriction

$$\pi_{\mathcal{W}}|_{\partial\mathcal{W}_{\text{reg}}} : \partial\mathcal{W}_{\text{reg}} \longrightarrow A_{\text{reg}} \quad (33)$$

are smooth submersions.

Proof. On the interior, $\pi_{\mathcal{W}}$ is the restriction of the product projection and is therefore a submersion.

At a boundary point (a, x) , a tangent vector (v, w) lies in $T_{(a,x)}\partial\mathcal{W}_{\text{reg}}$ exactly when

$$d_a F(v) + d_x F(w) = 0. \quad (34)$$

Given any $v \in T_a A_{\text{reg}}$, the regular-value condition makes $d_x F$ surjective onto $\mathbb{R}$. Hence one may choose w satisfying this equation. Therefore

$$d\pi_{\mathcal{W}}(v, w) = v \quad (35)$$

is surjective on the boundary as well. $\square$

Theorem 3.9 (Local Triviality of Regular Observation Windows). *The projection*

$$\pi_{\mathcal{W}} : \mathcal{W}_{\text{reg}} \longrightarrow A_{\text{reg}} \quad (36)$$

is a locally trivial smooth fibre bundle.

Consequently, for every $a_0 \in A_{\text{reg}}$, there exists an open neighbourhood $U \ni a_0$ and a diffeomorphism

$$\Psi_U : \pi_{\mathcal{W}}^{-1}(U) \longrightarrow U \times B_{a_0} \quad (37)$$

such that

$$\text{pr}_U \circ \Psi_U = \pi_{\mathcal{W}}. \quad (38)$$

Proof. By Assumption 3.5, $\pi_{\mathcal{W}}$ is proper. By Lemma 3.8, it is a submersion on both the interior and boundary of $\mathcal{W}_{\text{reg}}$. The boundary-compatible form of Ehresmann's fibration theorem therefore implies local smooth triviality. $\square$

Corollary 3.10 (Local Constancy of Window Topology). *For every $a_0 \in A_{\text{reg}}$, all windows B_a with a in a sufficiently small neighbourhood of a_0 are diffeomorphic to B_{a_0} . In particular, their smooth topology is locally constant.*

Proof. Each B_a is a fibre of the local product $U \times B_{a_0}$ supplied by Theorem 3.9. $\square$

Corollary 3.11 (Smoothly Varying Window Diffeomorphisms). *For every $a_0 \in A_{\text{reg}}$, the local trivialisation determines a smoothly parameterised family of diffeomorphisms*

$$\psi_a : B_{a_0} \longrightarrow B_a, \quad a \in U, \quad (39)$$

with $\psi_{a_0} = \text{id}_{B_{a_0}}$ after normalising the trivialisation at a_0 .

3.3 Quotienting

Definition 10 (Restriction Map). For every regular window $B \subseteq M$, define

$$r_B : \Gamma(P, S) \longrightarrow \Gamma^\infty(B, S|_B), \quad r_B(s) := s|_B. \quad (40)$$

Definition 11 (Ignored-Information Kernel). The ignored-information kernel associated with B is

$$K_B := \ker r_B = \{s \in \Gamma(P, S) : s|_B = 0\}. \quad (41)$$

It contains precisely those global field variations that vanish inside the observation window.

Definition 12 (Observational Equivalence). For $s_1, s_2 \in \Gamma(P, S)$, define

$$s_1 \sim_B s_2 \iff r_B(s_1) = r_B(s_2) \iff s_1 - s_2 \in K_B. \quad (42)$$

Definition 13 (Quotient Fibre). The quotient fibre induced by B is

$$Q_B(S) := \Gamma(P, S)/K_B. \quad (43)$$

Its elements are the equivalence classes

$$[s]_B := s + K_B. \quad (44)$$

Lemma 3.12 (Closedness of the Ignored-Information Kernel). *For every regular window B , the subspace K_B is closed in $\Gamma(P, S)$.*

Proof. By Assumption 3.1, r_B is continuous and its codomain is Hausdorff. Therefore

$$K_B = r_B^{-1}(\{0\}) \quad (45)$$

is closed. $\square$

Assumption 3.13 (Extension Property). For every regular window B , the restriction map r_B admits a continuous linear right inverse

$$e_B : \Gamma^\infty(B, S|_B) \longrightarrow \Gamma(P, S) \quad (46)$$

satisfying

$$r_B \circ e_B = \text{id}_{\Gamma^\infty(B, S|_B)}. \quad (47)$$

Lemma 3.14 (Quotient–Restriction Isomorphism). *The restriction map induces a topological vector-space isomorphism*

$$\bar{r}_B : Q_B(S) \longrightarrow \Gamma^\infty(B, S|_B), \quad \bar{r}_B([s]_B) := s|_B. \quad (48)$$

Proof. The map is well-defined because equivalent representatives have the same restriction. Its kernel is trivial by the definition of K_B . It is surjective by Assumption 3.13. Its inverse is

$$\bar{r}_B^{-1}(\sigma) = [e_B(\sigma)]_B, \quad (49)$$

which is continuous and independent of the chosen extension modulo K_B . Thus $\bar{r}_B$ is a topological vector-space isomorphism. $\square$

Lemma 3.15 (Field Filtering–Quotienting Commutativity). *Let $I = [\phi, \bar{\phi}]$. The field filtration induces a well-defined map*

$$\bar{\Pi}_{I,B} : Q_B(S) \longrightarrow Q_B(S_I), \quad \bar{\Pi}_{I,B}([s]_B) := [\Pi_I(s)]_B. \quad (50)$$

Moreover,

$$\bar{\Pi}_{I,B} \circ q_B^S = q_B^{S_I} \circ \Pi_I, \quad (51)$$

where q_B^S and $q_B^{S_I}$ are the respective quotient maps.

Proof. If $[s_1]_B = [s_2]_B$, then

$$s_1|_B = s_2|_B. \quad (52)$$

Applying the fibrewise projection Π_I gives

$$\Pi_I(s_1)|_B = \Pi_I(s_2)|_B, \quad (53)$$

so $[\Pi_I(s_1)]_B = [\Pi_I(s_2)]_B$. Hence the induced map is well-defined. The commutative identity follows directly from its definition; equivalently, it follows from the universal property of the quotient. $\square$

Theorem 3.16 (Local Triviality of the Quotient Fibres). *Fix $a_0 \in A_{\text{reg}}$. Let $U \ni a_0$ and*

$$\psi_a : B_{a_0} \longrightarrow B_a \quad (54)$$

be supplied by Corollary 3.11. Under Assumption 3.1, the family

$$\{Q_{B_a}(S)\}_{a \in U} \quad (55)$$

is smoothly and fibrewise linearly identifiable with $U \times Q_{B_{a_0}}(S)$.

Proof. Choose smoothly varying bundle isomorphisms

$$\tilde{\psi}_a : S|_{B_{a_0}} \longrightarrow S|_{B_a} \quad (56)$$

covering ψ_a . These induce pullback isomorphisms

$$T_a : \Gamma^\infty(B_a, S|_{B_a}) \longrightarrow \Gamma^\infty(B_{a_0}, S|_{B_{a_0}}). \quad (57)$$

Using Lemma 3.14, define

$$\tau_a := \bar{r}_{B_{a_0}}^{-1} \circ T_a \circ \bar{r}_{B_a} : Q_{B_a}(S) \longrightarrow Q_{B_{a_0}}(S). \quad (58)$$

Then

$$\Phi_U : \coprod_{a \in U} Q_{B_a}(S) \longrightarrow U \times Q_{B_{a_0}}(S), \quad (59)$$

$$\Phi_U(a, q) := (a, \tau_a q) \quad (60)$$

is a smoothly parameterised fibrewise linear isomorphism, with smooth inverse. $\square$

Corollary 3.17 (Smooth Construction of the Regular s -Catalogue). *The disjoint union*

$$\coprod_{a \in A_{\text{reg}}} Q_{B_a}(S) \quad (61)$$

admits a smooth locally trivial vector-bundle structure over A_{reg} .

Proof. The maps Φ_U from Theorem 3.16 form a bundle atlas. On overlapping neighbourhoods, their transition maps are smooth and fibrewise linear by Assumption 3.1. $\square$

4 Smooth Vectorisation

4.1 Constructing the s -Catalogue

Definition 14 (Regular s -Catalogue). The regular s -catalogue is

$$E_S^{\text{reg}} := \coprod_{a \in A_{\text{reg}}} (\{a\} \times Q_{B_a}(S)), \quad (62)$$

equipped with the projection

$$\pi_E : E_S^{\text{reg}} \longrightarrow A_{\text{reg}}, \quad \pi_E(a, q) = a. \quad (63)$$

Theorem 4.1 (Smooth Vector-Bundle Structure of the Regular s -Catalogue). *The projection*

$$\pi_E : E_S^{\text{reg}} \longrightarrow A_{\text{reg}} \quad (64)$$

is a smooth locally trivial vector bundle. For each $a_0 \in A_{\text{reg}}$, its local model is

$$U \times Q_{B_{a_0}}(S). \quad (65)$$

Proof. This is precisely the vector-bundle structure constructed in Corollary 3.17. The local trivialising maps are the maps Φ_U constructed in Theorem 3.16. $\square$

Corollary 4.2 (Local Smooth Representation of Regular Scenes). *Every point*

$$(a, [s]_{B_a}) \in E_S^{\text{reg}} \quad (66)$$

has a neighbourhood represented smoothly in

$$\mathbb{R}^{\dim A_{\text{reg}}} \oplus Q_{B_{a_0}}(S) \quad (67)$$

for some reference parameter a_0 .

More explicitly, if

$$\chi : U \longrightarrow \mathbb{R}^{\dim A_{\text{reg}}} \quad (68)$$

is a coordinate chart on the base, then

$$(a, [s]_{B_a}) \longmapsto (\chi(a), \tau_a([s]_{B_a})) \quad (69)$$

is a local smooth representation of the observed scene.

Proof. Compose the bundle trivialisation Φ_U with the base chart χ . Both maps are smooth and locally invertible. $\square$

4.2 Bundle Trivialisation.

Definition 15 (Local Bundle Trivialisation). Let

$$\pi_E : E_S^{\text{reg}} \longrightarrow A_{\text{reg}} \quad (70)$$

be the regular s -catalogue bundle. A local bundle trivialisation around $a_0 \in A_{\text{reg}}$ is a pair (U, τ_U) , where $U \subseteq A_{\text{reg}}$ is an open neighbourhood of a_0 and

$$\tau_U : \pi_E^{-1}(U) \longrightarrow U \times Q_{B_{a_0}}(S) \quad (71)$$

is a diffeomorphism satisfying

$$\text{pr}_U \circ \tau_U = \pi_E \quad (72)$$

and whose restriction to each fibre is linear.

Theorem 4.3 (Local Semantic Vectorisation). *Let (U, τ_U) be a local bundle trivialisation around $a_0 \in A_{\text{reg}}$, and let*

$$\chi_U : U \longrightarrow \chi_U(U) \subseteq \mathbb{R}^{\dim A_{\text{reg}}} \quad (73)$$

be a smooth coordinate chart. Then

$$h_U := (\chi_U \times \text{id}) \circ \tau_U, \quad (74)$$

$$h_U : \pi_E^{-1}(U) \longrightarrow V_{\text{int},U} := \chi_U(U) \times Q_{B_{a_0}}(S) \quad (75)$$

is a diffeomorphism.

Moreover, $V_{\text{int},U}$ is an open subset of the modelling vector space

$$\mathbb{V}_{\text{int},U} := \mathbb{R}^{\dim A_{\text{reg}}} \oplus Q_{B_{a_0}}(S). \quad (76)$$

Proof. The bundle trivialisation τ_U is a diffeomorphism by definition, and $\chi_U \times \text{id}$ is a diffeomorphism from $U \times Q_{B_{a_0}}(S)$ onto $\chi_U(U) \times Q_{B_{a_0}}(S)$. Their composition is therefore a diffeomorphism. Since $\chi_U(U)$ is open in $\mathbb{R}^{\dim A_{\text{reg}}}$, the image $V_{\text{int},U}$ is open in $\mathbb{V}_{\text{int},U}$. $\square$

Corollary 4.4 (Local Recovery of the Observed Equivalence Class). *For every*

$$v = h_U(a, [s]_{B_a}) \in V_{\text{int},U}, \quad (77)$$

the inverse map

$$h_U^{-1} : V_{\text{int},U} \longrightarrow \pi_E^{-1}(U) \quad (78)$$

recovers the unique pair

$$h_U^{-1}(v) = (a, [s]_{B_a}). \quad (79)$$

Proof. The claim follows from the bijectivity of the diffeomorphism h_U . The recovered object is the quotient class $[s]_{B_a}$, not a unique global representative s . $\square$

Corollary 4.5 (Uniqueness up to Smooth Coordinate Transformation). *Let*

$$h_U : \pi_E^{-1}(U) \longrightarrow V_{\text{int},U}, \quad h_V : \pi_E^{-1}(V) \longrightarrow V_{\text{int},V} \quad (80)$$

be two local semantic vectorisations. On every non-empty overlap,

$$h_V \circ h_U^{-1} \quad (81)$$

is a smooth diffeomorphism between the corresponding coordinate domains. Its action on each quotient fibre is linear.

Proof. The transition map is a composition of diffeomorphisms and is therefore a diffeomorphism. Fibrewise linearity follows from the fibrewise linearity of the bundle trivialisations. $\square$

Conjecture 4.6 (Global Trivialisation). *Suppose that A_{reg} is paracompact and smoothly contractible, that E_S^{reg} is a numerable smooth vector bundle, and that its Fréchet structure group admits the required smooth homotopy-classification property. Then there exists a global smooth fibrewise-linear isomorphism*

$$E_S^{\text{reg}} \cong A_{\text{reg}} \times Q_{B_{a_0}}(S) \quad (82)$$

for any reference point $a_0 \in A_{\text{reg}}$.

4.3 The Intrinsic Semantic Space

Definition 16 (Intrinsic Semantic Space). The intrinsic semantic space is the regular s -catalogue equipped with its local semantic-vectorisation atlas:

$$V_{\text{int}} := (E_S^{\text{reg}}, \{(V_{\text{int},U}, h_U)\}_U). \quad (83)$$

A point of V_{int} represents a regular observation parameter together with an observed physical equivalence class:

$$v_{\text{int}} \longleftrightarrow (a, [s]_{B_a}). \quad (84)$$

Unless Conjecture 4.6 holds, V_{int} is not claimed to be one global vector space.

Proposition 4.7 (Coordinate Covariance of the Intrinsic Space). *Let*

$$\varphi : M \longrightarrow M' \quad (85)$$

be a diffeomorphism and let

$$\tilde{\varphi} : S \longrightarrow S' \quad (86)$$

be a vector-bundle isomorphism covering φ . Define

$$\varphi_{\#} s := \tilde{\varphi} \circ s \circ \varphi^{-1}. \quad (87)$$

Then

$$s_1 \sim_B s_2 \iff \varphi_{\#} s_1 \sim_{\varphi(B)} \varphi_{\#} s_2. \quad (88)$$

Consequently,

$$\varphi_{\#}^V : V_{\text{int}}(P, S) \longrightarrow V_{\text{int}}(P', S') \quad (89)$$

defined by

$$\varphi_{\#}^V(a, [s]_{B_a}) := (a, [\varphi_{\#} s]_{\varphi(B_a)}) \quad (90)$$

is a smooth bundle isomorphism.

Proof. Equality of s_1 and s_2 on B is equivalent, under the invertible bundle map $\tilde{\varphi}$, to equality of their transformed sections on $\varphi(B)$. The induced quotient map is therefore well-defined and invertible. Smoothness follows from Assumption 3.1. $\square$

Proposition 4.8 (Compatibility across Field Filtrations). *For every retained field range $I = [\phi, \bar{\phi}]$, the fibrewise maps from Lemma 3.15 assemble into a smooth vector-bundle morphism*

$$\Pi_I^V : V_{\text{int}}(S) \longrightarrow V_{\text{int}}(S_I) \quad (91)$$

defined by

$$\Pi_I^V(a, [s]_{B_a}) := (a, [\Pi_I(s)]_{B_a}). \quad (92)$$

If $I_2 \subseteq I_1$, then

$$\Pi_{I_1 \rightarrow I_2}^V \circ \Pi_{I_1}^V = \Pi_{I_2}^V. \quad (93)$$

Proof. Well-definedness and fibrewise linearity follow from Lemma 3.15. Smoothness follows in each bundle chart from the smoothness of the field projections. The composition identity follows from Lemma 3.4. $\square$

Corollary 4.9 (Coordinate Agnosticism). *A change of physical coordinates changes the coordinate representative of V_{int} but not its smooth isomorphism class.*

Corollary 4.10 (Scale Dependence). *The intrinsic semantic space depends on the descriptive specification:*

$$V_{\text{int}} = V_{\text{int}}(\varepsilon, \mathcal{N}, [\phi, \bar{\phi}]). \quad (94)$$

Changing the resolution, normalisation, or retained field content may therefore change its fibres, dimension, topology, and subsequent categorisation. No canonical identification across scales is implied without an additional inter-scale map.

Corollary 4.11 (Recovery of the Restricted Scene). *For every $[s]_B \in Q_B(S)$, the quotient-restriction isomorphism recovers*

$$\bar{r}_B([s]_B) = s|_B. \quad (95)$$

It does not recover the values of s on $M \setminus B$.

Proof. If $k \in K_B$, then

$$(s + k)|_B = s|_B. \quad (96)$$

Thus every representative of $[s]_B$ has the same restriction to B , while representatives may differ arbitrarily outside B . $\square$

5 Observer Relative Geometry

5.1 The General Extrinsic Problem

Definition 17 (Complete Probe–Observer Physical Scene). A complete probe–observer physical scene is a tuple

$$\Sigma := (P, S, s, B_p, B_{\text{obs}}, y_0, \mathcal{P}, F_{\text{obs}}), \quad (97)$$

where $s \in \Gamma(P, S)$ is the complete physical configuration, B_p and B_{obs} mark the probe and observer regions, y_0 is the observer’s prepared state, $\mathcal{P}$ is the interaction protocol, and F_{obs} governs the observer’s physical response dynamics.

The corresponding intrinsic probe and observer states are

$$p = [s]_{B_p}, \quad o = [s]_{B_{\text{obs}}}. \quad (98)$$

Definition 18 (Observer-Response Space). The observer-response space is a smooth Hilbert manifold

$$(Y_{\text{obs}}, g_Y) \quad (99)$$

whose points represent physically distinguishable observer responses. For the Jacobian construction, work on a fixed-resolution Hilbert realisation

$$U_{\text{int}} \subseteq H_{\text{int}} \quad (100)$$

of a local intrinsic coordinate domain.

Assumption 5.1 (Observer-Relative Causal Meaning). The meaning of a probe relative to a fixed observer is wholly determined by the physical response that the probe induces in that observer under the fixed interaction protocol. Locally, semantic sensitivity to a probe perturbation u is therefore determined by the corresponding observer-response perturbation.

Assumption 5.2 (Physical Closure of the Response Law). The response dynamics F_{obs} , preparation y_0 , protocol $\mathcal{P}$, and readout used to construct the observer response are determined by the complete physical scene Σ . No target labels or externally specified semantic metric enter the response law.

Definition 19 (Observer-Response Map). For a fixed observer and protocol, the observer-response map is

$$R_{\text{obs}} : U_{\text{int}} \longrightarrow Y_{\text{obs}}, \quad p \longmapsto R_{\text{obs}}(p), \quad (101)$$

where $R_{\text{obs}}(p)$ is the observer's settled or otherwise protocol-specified physical response to the probe p .

5.2 Relative Causal Meaning.

Assumption 5.3 (Smooth Observer Dynamics). For a fixed probe p , the observer state satisfies

$$\dot{y}(t) = F_{\text{obs}}(y(t), p), \quad y(0) = y_0, \quad (102)$$

where F_{obs} is of class C^r , with $r \geq 1$, and generates a unique response trajectory over the prescribed protocol.

Assumption 5.4 (Fixed Preparation and Interaction Protocol). The initial observer state y_0 , probe-presentation rule, interaction duration, coupling law, and response readout are held fixed while different probes are compared.

Assumption 5.5 (Regular or Constant-Rank Response). The map R_{obs} is of class C^r , and its derivative has constant rank on each regular response region. In the Hilbert setting, its kernel and range are assumed closed and complemented. Define the regular immersion region by

$$U_{\text{imm}} := \{p \in U_{\text{int}} : \ker DR_{\text{obs}}|_p = \{0\}\}. \quad (103)$$

Lemma 5.6 (Structural Jacobian from Observer Dynamics). *Suppose that the settled response $y_{\infty}(p)$ is determined by*

$$F_{\text{obs}}(y_{\infty}(p), p) = 0 \quad (104)$$

and that

$$D_y F_{\text{obs}}|_{(y_{\infty}(p), p)} \quad (105)$$

is a bounded linear isomorphism. Then y_{∞} is locally C^r and

$$D_p y_{\infty}(p) = - \left(D_y F_{\text{obs}}|_{(y_{\infty}(p), p)} \right)^{-1} D_p F_{\text{obs}}|_{(y_{\infty}(p), p)}. \quad (106)$$

If $R_{\text{obs}} = L_{\text{obs}} \circ y_{\infty}$ for a smooth physical readout L_{obs} , then

$$DR_{\text{obs}}|_p = DL_{\text{obs}}|_{y_{\infty}(p)} \circ D_p y_{\infty}(p). \quad (107)$$

Proof. Differentiate

$$F_{\text{obs}}(y_{\infty}(p), p) = 0 \quad (108)$$

in a probe direction u . The chain rule gives

$$D_y F_{\text{obs}}(D_p y_{\infty}(p)u) + D_p F_{\text{obs}}(u) = 0. \quad (109)$$

Invertibility of $D_y F_{\text{obs}}$ yields the stated derivative. The final identity follows by differentiating $R_{\text{obs}} = L_{\text{obs}} \circ y_{\infty}$. $\square$

Corollary 5.7 (Observer Dependence of Semantic Sensitivity). *Two observers governed by different physical response laws may induce different derivatives*

$$DR_{\text{obs}}^{(1)}|_p \neq DR_{\text{obs}}^{(2)}|_p \quad (110)$$

for the same probe p . Consequently, semantic sensitivity is observer-relative rather than a function of the probe alone.

5.3 Metric Curvature via Jacobian Probing.

Definition 20 (Structural Jacobian). The structural Jacobian at $p \in U_{\text{int}}$ is

$$J_{\text{obs}}(p) := DR_{\text{obs}}|_p : T_p V_{\text{int}} \longrightarrow T_{R_{\text{obs}}(p)} Y_{\text{obs}}. \quad (111)$$

Definition 21 (Semantic Warping Operator). Equip the fixed-resolution intrinsic tangent model with a reference Hilbert inner product. Let $J_{\text{obs}}(p)^*$ denote the adjoint with respect to this reference inner product and g_Y . Define

$$\Omega_{\text{obs}}(p) := J_{\text{obs}}(p)^* J_{\text{obs}}(p). \quad (112)$$

The reference inner product is used only to represent the pulled-back bilinear form as an operator; it is not added to the observer-relative metric.

Theorem 5.8 (Observer-Relative Pullback Metric). *The pullback of g_Y through R_{obs} is*

$$g_{\text{obs}} := R_{\text{obs}}^* g_Y. \quad (113)$$

For $u, w \in T_p V_{\text{int}}$,

$$g_{\text{obs},p}(u, w) = g_{Y, R_{\text{obs}}(p)}(J_{\text{obs}}(p)u, J_{\text{obs}}(p)w) \quad (114)$$

$$= \langle u, \Omega_{\text{obs}}(p)w \rangle_{\text{int}}. \quad (115)$$

Proof. The first equality is the definition of the pullback of a metric. By the defining property of the adjoint,

$$\langle J_{\text{obs}}u, J_{\text{obs}}w \rangle_Y = \langle u, J_{\text{obs}}^* J_{\text{obs}}w \rangle_{\text{int}}. \quad (116)$$

Substituting $\Omega_{\text{obs}} = J_{\text{obs}}^* J_{\text{obs}}$ gives the result. $\square$

Corollary 5.9 (Positive-Definiteness on the Immersion Region). *For every $p \in U_{\text{imm}}$ and every non-zero $u \in T_p V_{\text{int}}$,*

$$g_{\text{obs},p}(u, u) = \|J_{\text{obs}}(p)u\|_Y^2 > 0. \quad (117)$$

Hence g_{obs} is positive-definite on U_{imm} .

Corollary 5.10 (Degeneracy along Observer-Insensitive Directions). *For every $u \in T_p V_{\text{int}}$,*

$$g_{\text{obs},p}(u, u) = 0 \iff J_{\text{obs}}(p)u = 0 \iff u \in \ker \Omega_{\text{obs}}(p). \quad (118)$$

Thus the pullback form is degenerate precisely along directions to which the observer is locally insensitive.

Corollary 5.11 (Finite-Dimensional Topology Preservation). *Suppose that $V_{\text{int}}^{(n)}$ is a finite-dimensional truncation and that $J_{\text{obs}}(p)$ is injective throughout a neighbourhood U . Then g_{obs} and any smooth positive-definite reference metric on U induce the same topology.*

Proof. At each $p \in U$, the two positive-definite quadratic forms are equivalent. Locally, there exist constants

$$0 < c_p \leq C_p < \infty \quad (119)$$

such that

$$c_p g_0(u, u) \leq g_{\text{obs}}(u, u) \leq C_p g_0(u, u). \quad (120)$$

Their induced norms therefore generate the same local open sets. $\square$

[Infinite-Dimensional Limit] Injectivity of J_{obs} in an infinite-dimensional Hilbert space does not imply a uniform lower bound

$$\|J_{\text{obs}}u\|_Y \geq c\|u\|_{\text{int}} \quad (121)$$

for some $c > 0$. Consequently, asymptotically vanishing metric weights may induce a strictly weaker topology. No general infinite-dimensional topology-preservation result is claimed.

Lemma 5.12 (Self-Adjoint Positive-Semidefinite Spectral Structure). *For every p , the operator*

$$\Omega_{\text{obs}}(p) = J_{\text{obs}}(p)^* J_{\text{obs}}(p) \quad (122)$$

is bounded, self-adjoint, and positive-semidefinite. Its spectrum satisfies

$$\sigma(\Omega_{\text{obs}}(p)) \subseteq [0, \infty). \quad (123)$$

In finite dimensions it admits an orthonormal eigenbasis with non-negative eigenvalues. The same discrete eigenbasis conclusion in infinite dimensions requires additional compactness or discrete-spectrum conditions.

Proof. Self-adjointness follows from

$$(J^* J)^* = J^* J. \quad (124)$$

For every u ,

$$\langle u, J^* J u \rangle = \|Ju\|^2 \geq 0. \quad (125)$$

The spectral inclusion follows from positivity and the spectral theorem. $\square$

Lemma 5.13 (Smooth Spectral Projectors under a Spectral Gap). *Let $p \mapsto \Omega_{\text{obs}}(p)$ be C^r in operator norm. Suppose that, throughout an open set U , a spectral cluster is separated from the remaining spectrum by a fixed closed contour γ in the resolvent set. Then the associated Riesz projector*

$$P_\gamma(p) := \frac{1}{2\pi i} \int_\gamma (zI - \Omega_{\text{obs}}(p))^{-1} dz \quad (126)$$

is of class C^r on U .

The spectral subspace therefore varies smoothly even when individual eigenvectors within the isolated cluster are non-unique.

Proof. The resolvent is C^r in p wherever γ remains inside the resolvent set. Differentiation under the contour integral therefore shows that P_γ is C^r . $\square$

Conjecture 5.14 (Degeneracy-Crossing Semantic Transition). *Let $p(t)$ be a smooth path of probes and let $\delta_{\text{spec}}(p)$ denote the spectral gap separating a designated observer-response subspace from its complement. Under the response-equivalence construction of Section 5.2, a change of observer-relative category may occur only at a parameter t_* for which*

$$\delta_{\text{spec}}(p(t_*)) = 0. \quad (127)$$

The converse—that every such gap closure produces a category transition—is not asserted without an additional dynamical bifurcation condition.

[Exogenous Two-Field Observer] Let

$$H_{\text{int}} = H_{\text{obs}} = \mathbb{R}^2 \quad (128)$$

and stipulate

$$R_{\text{obs}}(q) = \begin{bmatrix} 2 & 0 \\ 0 & \varepsilon \end{bmatrix} q, \quad 0 < \varepsilon \ll 1. \quad (129)$$

Then

$$J_{\text{obs}} = \begin{bmatrix} 2 & 0 \\ 0 & \varepsilon \end{bmatrix}, \quad \Omega_{\text{obs}} = \begin{bmatrix} 4 & 0 \\ 0 & \varepsilon^2 \end{bmatrix}. \quad (130)$$

Thus the first direction is stretched while the second is compressed. This example stipulates rather than derives the observer sensitivity.

[Minimal Dynamical Observer] Let the observer satisfy

$$\dot{y} = -\Lambda y + Kq, \quad (131)$$

where

$$\Lambda = \begin{bmatrix} \lambda_0 & 0 \\ 0 & \lambda_1 \end{bmatrix}, \quad K = \begin{bmatrix} k_0 & 0 \\ 0 & k_1 \end{bmatrix}, \quad \lambda_0, \lambda_1 > 0. \quad (132)$$

The unique stable equilibrium is

$$y_\infty(q) = \Lambda^{-1} Kq. \quad (133)$$

If

$$\frac{k_0}{\lambda_0} = 2, \quad \frac{k_1}{\lambda_1} = \varepsilon, \quad (134)$$

then

$$J_{\text{obs}} = D_q y_\infty = \begin{bmatrix} 2 & 0 \\ 0 & \varepsilon \end{bmatrix}, \quad \Omega_{\text{obs}} = \begin{bmatrix} 4 & 0 \\ 0 & \varepsilon^2 \end{bmatrix}. \quad (135)$$

Here the Jacobian follows from an explicit response law, although the gain and decay parameters remain physical assumptions.

[Independent Observer Assumptions] Let two observers act on the same probe space $\mathbb{R}^2$ through

$$R_A(q) = \begin{bmatrix} 2 & 0 \\ 0 & \varepsilon \end{bmatrix} q, \quad R_B(q) = \begin{bmatrix} \varepsilon & 0 \\ 0 & 2 \end{bmatrix} q. \quad (136)$$

Then

$$\Omega_A = \begin{bmatrix} 4 & 0 \\ 0 & \varepsilon^2 \end{bmatrix}, \quad \Omega_B = \begin{bmatrix} \varepsilon^2 & 0 \\ 0 & 4 \end{bmatrix}. \quad (137)$$

The same intrinsic probe geometry therefore supports opposite observer-relative sensitivities. The probe space alone does not select between them.

6 Observer Relative Categorisation

6.1 The General Categorisation Problem

Definition 22 (Partition Space). For any non-empty set V , define

$$\text{Part}(V) := \left\{ \mathcal{C} \subseteq 2^V : \emptyset \notin \mathcal{C}, \bigcup_{C \in \mathcal{C}} C = V, C \cap C' = \emptyset \text{ whenever } C \neq C' \right\}. \quad (138)$$

Definition 23 (Assumption-Conditioned Target Partition). Let Λ be an admissible observer-assumption object and let

$$\kappa_\Lambda^* : V_{\text{int}} \longrightarrow I_\Lambda \quad (139)$$

be its target category map. The corresponding target partition is

$$\mathcal{C}_\Lambda^* := \{ (\kappa_\Lambda^*)^{-1}(\{i\}) : i \in \text{im}(\kappa_\Lambda^*) \} \in \text{Part}(V_{\text{int}}). \quad (140)$$

Definition 24 (Geometric Discretiser). Let

$$X := (V_{\text{int}}, g_{\text{obs}}). \quad (141)$$

A geometric discretiser is a map

$$D_\mu : X \longrightarrow \text{Part}(V_{\text{int}}), \quad (142)$$

where μ records the chosen geometric construction and its parameters.

Theorem 6.1 (No Assumption-Free Recovery). *Suppose that the same X is compatible with two admissible assumption objects Λ_1 and Λ_2 satisfying*

$$\mathcal{C}_{\Lambda_1}^* \neq \mathcal{C}_{\Lambda_2}^*. \quad (143)$$

Then no discretiser depending only on X can guarantee recovery of $\mathcal{C}_\Lambda^$ for every admissible Λ .*

Proof. The input X is identical under Λ_1 and Λ_2 , so D_μ must return the same partition in both cases. One partition cannot equal two distinct target partitions. Hence the discretiser fails for at least one admissible assumption object. $\square$

beginncorollary[No Canonical Categorisation from Bare Geometry] Whenever distinct target partitions are compatible with the same bare physical geometry, that geometry does not canonically determine a categorisation.

Corollary 6.2 (Category Names up to Permutation). *FLet $\mathcal{C} \in \text{Part}(V_{\text{int}})$ and let*

$$\ell : \mathcal{C} \longrightarrow I \quad (144)$$

be any bijective naming map. For every permutation $\sigma : I \rightarrow I$, the map $\sigma \circ \ell$ assigns different names but induces the same partition. Hence the recovered category cells do not determine their conventional names.

6.2 Measure-based fibre quotienting.

Definition 25 (Observer-Response Dynamics). For each probe $p \in V_{\text{int}}$, let

$$\Phi_p^t(y_0) \quad (145)$$

denote the observer trajectory generated by

$$\dot{y} = F_{\text{obs}}(y, p), \quad y(0) = y_0. \quad (146)$$

Definition 26 (Settled Response and Invariant Response Measure). The empirical response measure at duration T is

$$\nu_{p,T} := \frac{1}{T} \int_0^T \delta_{\Phi_p^t(y_0)} dt. \quad (147)$$

When $\nu_{p,T}$ converges weakly as $T \rightarrow \infty$, denote its limit by μ_p . Define the settled response object by

$$z_p := (\mathcal{A}_p, \mu_p), \quad \mathcal{A}_p := \text{supp}(\mu_p). \quad (148)$$

For a stable equilibrium $y_\infty(p)$,

$$\mu_p = \delta_{y_\infty(p)}. \quad (149)$$

Assumption 6.3 (Existence and Uniqueness of Settled Responses). For every probe under consideration, the fixed preparation and protocol produce a unique invariant response measure μ_p and hence a unique settled response object z_p .

Assumption 6.4 (Observer-Relative Attractor Equivalence). The observer's physical response architecture induces an equivalence relation

$$\equiv_{\text{obs}} \quad (150)$$

on settled response objects. Two probes belong to the same observer-relative category exactly when their settled responses are equivalent under $\equiv_{\text{obs}}$. This equivalence criterion is physically instantiated by the observer rather than supplied by an external vocabulary.

Lemma 6.5 (Response Equivalence). *The relation $\sim_{\text{obs}}$ is an equivalence relation on V_{int} .*

Proof. Reflexivity, symmetry, and transitivity follow from the corresponding properties of $\equiv_{\text{obs}}$. $\square$

Theorem 6.6 (Observer-Relative Category Partition). *The collection of response-equivalence classes*

$$\mathcal{C}_{\text{obs}} := V_{\text{int}} / \sim_{\text{obs}} = \{[p]_{\text{obs}} : p \in V_{\text{int}}\} \quad (151)$$

is an element of $\text{Part}(V_{\text{int}})$.

Proof. Every probe belongs to its own equivalence class, so the classes cover V_{int} . Two equivalence classes are either equal or disjoint. Each class is non-empty. They therefore form a partition. $\square$

Theorem 6.7 (Local Stability of Categories). *For every*

$$p \in V_{\text{int}} \setminus \mathfrak{B}_{\text{obs}}, \quad (152)$$

there exists an open neighbourhood U_p such that

$$q \sim_{\text{obs}} p \quad \text{for every } q \in U_p. \quad (153)$$

Thus the category map is locally constant away from the response-bifurcation set.

Proof. By Assumption ??, the settled response of p persists throughout a sufficiently small neighbourhood without leaving its $\equiv_{\text{obs}}$ -equivalence class. Hence every q in that neighbourhood satisfies $q \sim_{\text{obs}} p$. $\square$

Corollary 6.8 (Location of Category Transitions). *Let $p : [0, 1] \rightarrow V_{\text{int}}$ be continuous. If $p(0)$ and $p(1)$ occupy different observer-relative categories, then*

$$p([0, 1]) \cap \mathfrak{B}_{\text{obs}} \neq \emptyset. \quad (154)$$

Corollary 6.9 (Spectral Identification of Category Transitions). *Define the relevant spectral-gap closure set by*

$$\mathfrak{Z}_{\text{spec}} := \{p \in V_{\text{int}} : \delta_{\text{spec}}(p) = 0\}. \quad (155)$$

If

$$\mathfrak{B}_{\text{obs}} \subseteq \mathfrak{Z}_{\text{spec}}, \quad (156)$$

then every observer-relative category transition occurs at a spectral degeneracy. If equality holds, the spectral degeneracies identify the complete response-bifurcation set.

7 Conclusion

At a fixed descriptive scale and on the regular region of the construction, a physically closed probe–observer scene determines a locally smooth representation of its observed physical equivalence classes. The observer’s physical response law induces a pullback geometry, while equivalence among stable response regimes induces an unlabelled category partition. The resulting categories are observer-relative response categories rather than conventional names; agreement with ordinary human concepts remains an empirical conjecture.

The result is local rather than universally global. It may fail at irregular observation windows, topology-changing window events, non-extendable restrictions, non-regular observer responses, response-bifurcation points, or an uncontrolled infinite-dimensional limit. Global trivialisation, spectral identification of all category transitions, agreement with conventional human categories, and stable infinite-dimensional recovery remain open.

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